Software Requirements Specification

for

Swifty

Version 1.0

Prepared by

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Course: CS253

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Harsh, Pulkit, Diya, Vanshika, Sreeja, Shruti, Saurya, Vasu, Divya, Rishitesh	Final draft of the SRS document after resolving all contentions in the features, and editing initial drafts.	21/1/2026

1.Introduction

1.1 Product Scope

The Student Gymkhana at IIT Kanpur acts as the central hub for student activities, managing numerous clubs, societies, and hobby groups. Currently, the administrative backbone of these bodies relies heavily on a manual, paper-based workflow. At present, the workflow for obtaining approvals from the Dean of Student Affairs (DOSA) office is entirely offline. Club coordinators are required to manually draft formal letters, obtain physical signatures from multiple authorities such as the Council Head, President Students' Gymkhana (PSG), and the Associate Dean of Student Affairs (ADSA), and submit the documents in person. This process is highly time-consuming, error-prone, and inefficient, often leading to significant delays due to the unavailability of signatories and repeated redrafting of documents.

Swifty (Gymkhana Automation System) is an envisioned web-based application designed to digitize and streamline this entire administrative lifecycle. The application aims to replace the physical "run-around" with a centralized digital dashboard where coordinators can raise service requests such as permission letters, venue bookings, and bill reimbursement directly online.To facilitate seamless event planning, the system will include an **interactive venue scheduler**: by simply selecting a specific date and time, coordinators can view the real-time availability for all lecture halls and campus venues, ensuring zero booking conflicts.The system will feature a robust hierarchical approval workflow, automatically routing digital forms to the correct authorities (PSG, Faculty Advisor, DOSA officials) for instant review and digital signature via OTP.

Crucially, the application empowers Club Coordinators with a dedicated personal account. On their home page, coordinators will have immediate visibility of their club's **current budget and expenditures**, allowing for better financial planning. This dashboard also acts as a digital archive, enabling them to view **past proposals and letters submitted** previously, eliminating the need to maintain physical files. By combining financial oversight with real-time status tracking, Swifty aims to eliminate administrative friction, allowing students to focus on innovation and events rather than bureaucracy.

1.2 Intended Audience and Document Overview

This software requirement documentation is written to be used and understood by:

- **Software Development Team and Individual Contributors** that will design and construct this software as per the requirements of a particular institution, in this case the **Students' Gymkhana** and **DOSA (Dean of Student Affairs) office**, Indian Institute of Technology Kanpur.
- **Product Managers**, in this case the **Teaching Assistants and Course Instructor**, whose responsibility is to overlook the planning, development and execution of the software development.
- **Testers**, who perform product and quality checks and carry out extensive testing and perform a quality check on the designed software and give their feedback on the interface, areas of improvement, etc.

- **End-Users**, which includes:
    - **Administrative Authorities:** The **President, Student Gymkhana (PSG)**, **General Secretaries (GenSec)**, **Faculty Advisors**, and DOSA officials who grant approvals.
    - **Club Representatives: Club Coordinators** and Executives responsible for raising requests and managing club finances.
    - **General Student Body:** Students who can access the platform for **public viewing of upcoming events** without the need to log in.

1.3 Definitions, Acronyms and Abbreviations

- **ADSA** Associate Dean of Student Affairs
- **API** Application Programming Interface
- **CSV** Comma-Separated Values
- **DOSA** Dean of Student Affairs
- **FacAd** Faculty Advisor
- **GenSec** General Secretary
- **LDAP** Lightweight Directory Access Protocol
- **LHC** Lecture Hall Complex
- **MoU** Memorandum of Understanding
- **OTP** One-Time Password
- **PSG** President, Student Gymkhana
- **SRS** Software Requirement Specification
- **UI** User Interface

1.4 Document Conventions

To ensure clarity and ease of navigation throughout this document, the following formatting standards have been applied:

- **Section Headings:** Primary headings are distinguished by **bold, text in Arial (font size 20)**.
- **Subsection Headings:** Secondary headings utilize **bold Arial (font size 14)** to maintain consistency with the content style.
- **Body Text:** The main content of the document is rendered in **Arial (font size 11)**.
- **Emphasis:** Significant terms and abbreviations are highlighted using **bold text** to facilitate quick scanning.
- **Alignment and Structure:** The entire text is **justified**. Indentation is actively employed to visually organize and display the hierarchy of information.
- **Standards:** This document primarily adheres to **IEEE formatting**, numbering, and indentation guidelines. Any deviations from the IEEE standard are explicitly specified.

1.5 References and Acknowledgments

References:

This document has been structured and formatted in accordance with the **IEEE Software Requirements Specification Template**, which served as the primary style guide.

Acknowledgment:

The core architectural principles and system design strategies implemented in this project were established through the lecture notes and supervision of our Course Instructor, **Dr. Indranil Saha**. We also explicitly acknowledge the invaluable guidance and mentorship provided by our Teaching Assistant, **Shaikh Imran**, throughout the development process.

2. Overall Description

2.1 Product Overview

The motivation behind this project is to revolutionize the administrative efficiency and transparency of the Student Gymkhana at IIT Kanpur. To solve the persistent bottlenecks caused by the manual, paper-based workflow such as the unavailability of signatories, opaque approval statuses, and frequent loss of documents, we plan on implementing an advanced **Gymkhana Automation System (Swifty)**. The application will serve as a centralized digital platform that replaces the physical routing of files with a secure, hierarchical approval workflow.

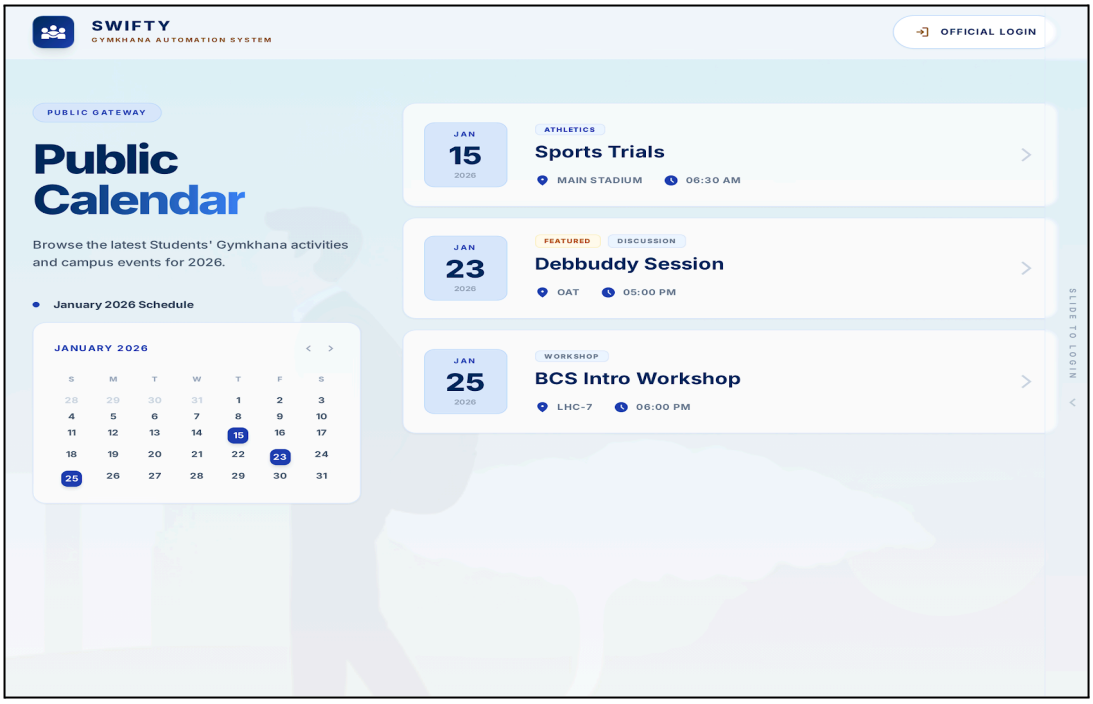
The system will feature a robust **Service Request Module** where club coordinators can draft and submit proposals for events, permission letters, and bill reimbursements. We plan on implementing an intelligent **Interactive Venue Scheduler** that displays the real-time availability of lecture halls and campus venues, allowing coordinators to select slots without conflict. Crucially, the platform will include a **Financial Dashboard** that provides coordinators with a live view of their club's budget utilization, expenditure history, and past proposals, ensuring better financial planning.

For the approving authorities (PSG, Faculty Advisors, ADSA), the system offers a **Verification Portal** where they can review documents and grant approvals via digital signatures (OTP-based) instantly, regardless of their physical location. The client (Student Gymkhana and DOSA Office) will have the option to view holistic data regarding the number of events conducted, budget processing times, and club activity levels to make well-informed managerial decisions. Additionally, the system will feature a **Public Event Calendar** to keep the general student body informed about upcoming campus activities.

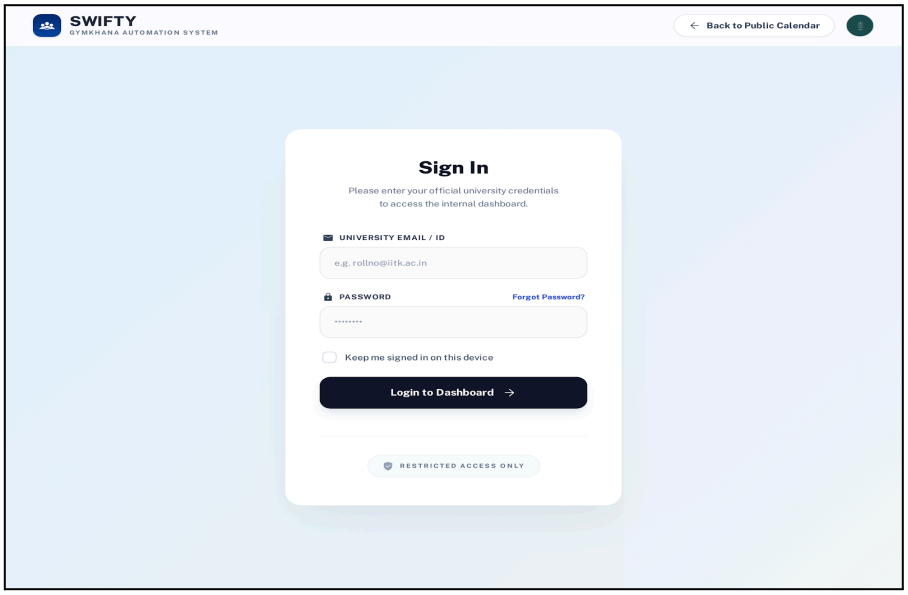
2.2 Product Functionality

2.2.1. USER LOGIN + PUBLIC CALENDAR

- **Public Calendar:** The homepage features a live calendar for the public to track upcoming events and for Coordinators to instantly see which venues are already booked.



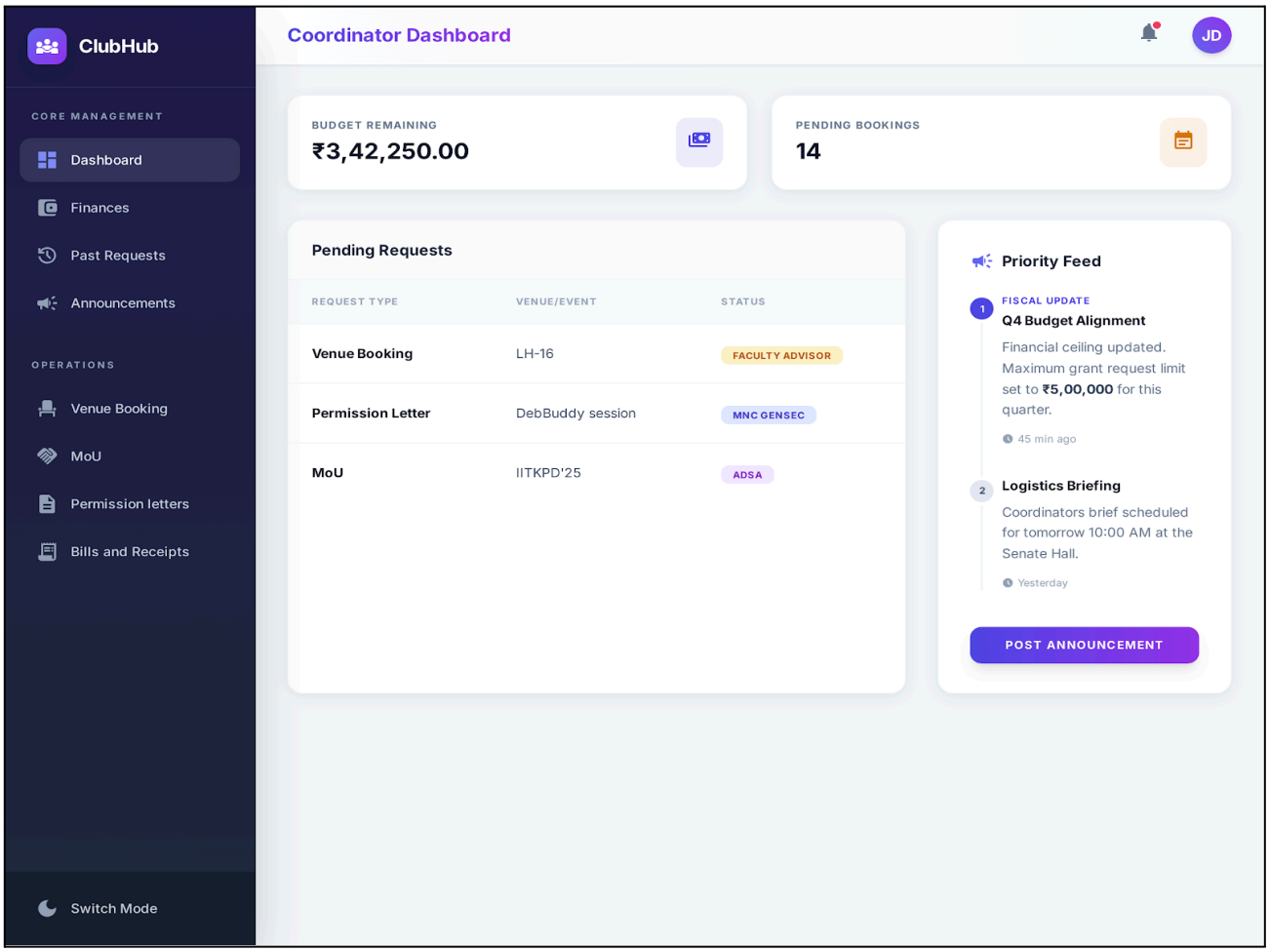
- **Login:** Authorized users can log in securely using their IITK email ID and a custom password to access their Dashboards to manage their respective tasks and approvals.



2.2.2. COORDINATOR DASHBOARD

Displays real-time **remaining budget**, **pending requests**, and active notifications. The sidebar menu contains the following options:

- **Permission Letter:** A digital form with automated hierarchical routing (**GenSec** → **FacAd** → **ADSA**) and real-time status tracking.
- **Venue Booking:** Requests institute locations and tracks approvals.
- **MoU Management:** Facilitates the submission and tracking of Memorandums of Understanding for external guests and organizations.
- **Financial Management:** Tools to view budget balances, process reimbursements, and maintain a detailed transaction log for auditing.
- **Bills & Receipts:** A repository for uploading documentation of bills and receipts for approval.
- **Announcements & History:** Sections to view official notices and a searchable archive of all past requests.

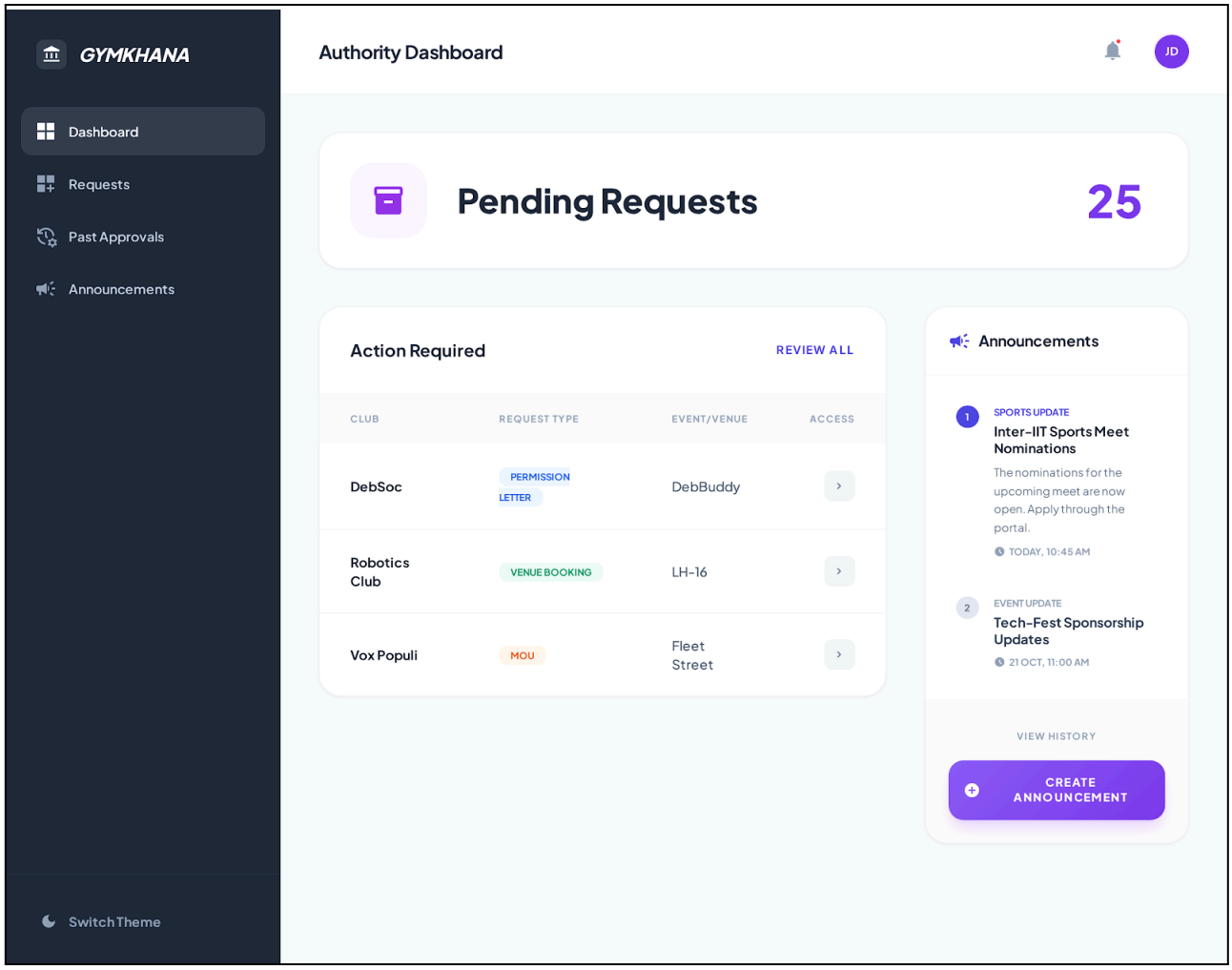


2.2.3. AUTHORITY DASHBOARD

- The Dashboard for the Authorities will display the number of pending requests, few details of the requests in tabular form and an announcements tab.

- The menu will have the following options:

    1. **Dashboard**: Navigates to the authority dashboard
    2. **Requests:** It will list down the current requests in the same format as they are displayed on the dashboard and they can be expanded
    3. **Past Approvals:** Will list down the past approvals
    4. **Announcements:** It will expand the announcements tab, and provide a button to create a new announcements



2.3 Design and Implementation Constraints

- The document database is an integral part of the software, and its storage capacity needs to be sufficient for archiving permission letters and bills.
- The server must have enough compute power and resources to accommodate concurrent requests from multiple clubs during peak event seasons without delay.
- The system relies on the institute's email gateway for OTP generation; any downtime in the email service may temporarily affect the approval workflow.
- Users with lower-level privileges (e.g., Coordinators) must not be able to access data or approval functions reserved for higher authorities (e.g., DOSA or GenSec).
- The client-side interface must be fully functional on all modern web browsers to ensure accessibility for administrative officials on varying devices.

2.4 Assumptions and Dependencies

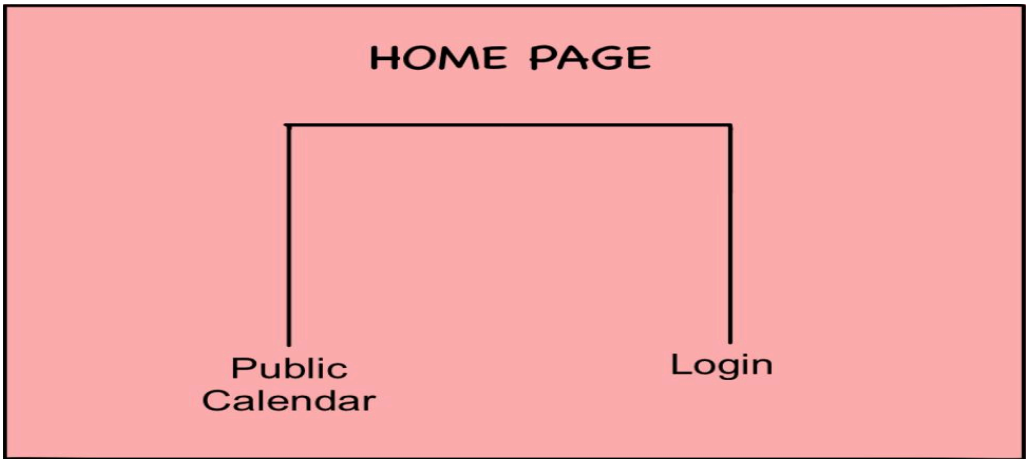
- The system is designed to handle a maximum of 150 registered users, encompassing Club Coordinators and Authorities.A key operational assumption is the timely processing of pending requests (acceptance or decline) by the approving authorities. Efficient workflow hinges on prompt action from authorities to avoid processing delays.
- It is assumed that club coordinators will uphold the institute's honor code in all their interactions with the system. Specifically, the system is built on the premise that only genuine bills will be submitted for reimbursement through the portal, and no fraudulent claims will be made.
- Venue verification functionality relies on a secure ID and OTP mechanism. The system assumes that the designated security guard or venue in-charge possesses the necessary equipment or protocol to successfully validate the unique ID and OTP presented by the user.
- The automated email notifications sent by the application are assumed to operate unimpeded by the Institute's firewall settings.
- The system operates with the understanding that academic lecture schedules and student event bookings will not conflict. This is based on the assumption that the database managing available slots for lecture halls has already been meticulously updated to exclude all academic class timings.
- It is assumed that once a request (venue booking or permission) receives final approval from the authority, the initiator cannot unilaterally cancel or withdraw it through the portal; such cancellations will require administrative intervention.

3. Specific Requirements

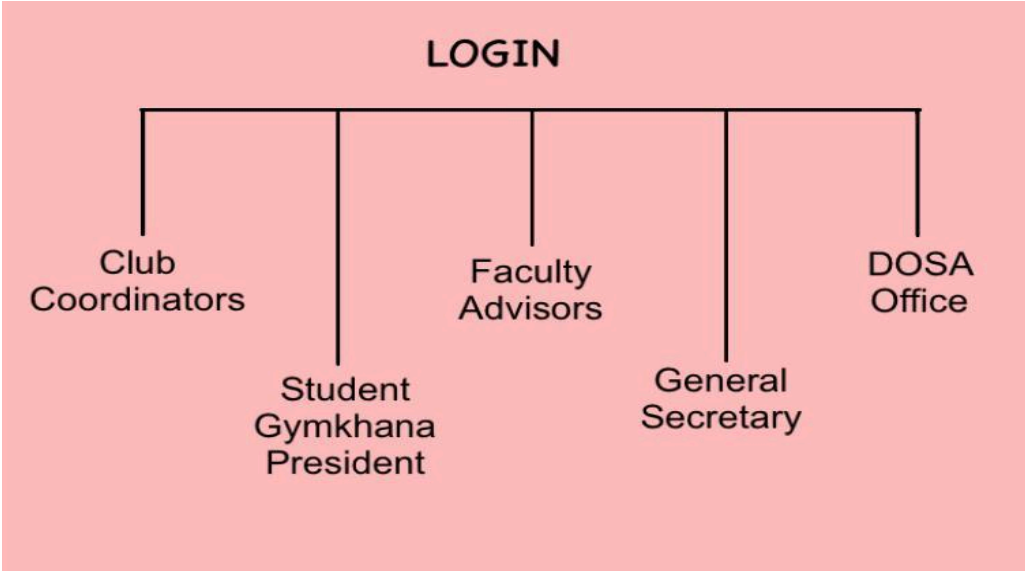
3.1 External Interface Requirements

3.1.1. User Interfaces

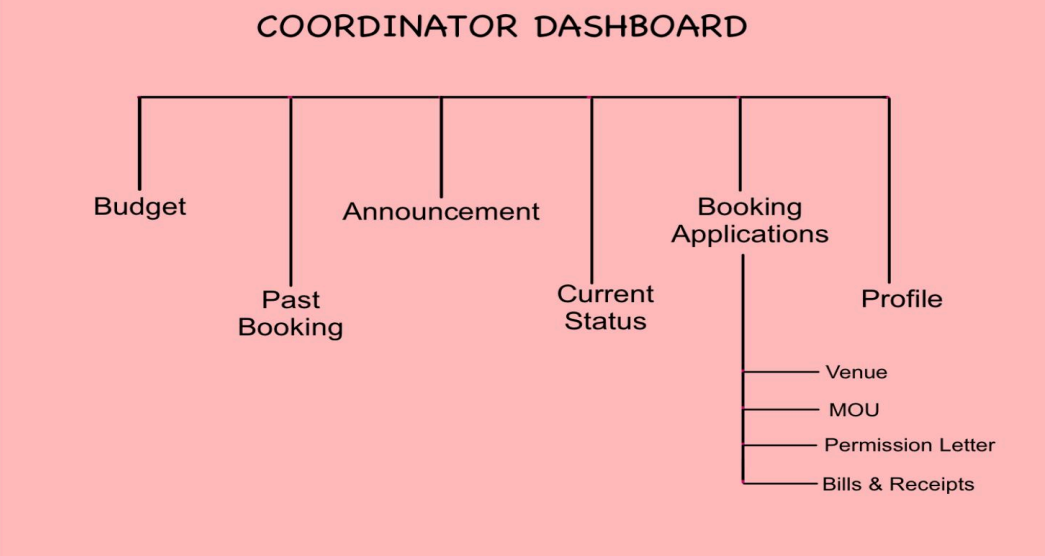
The homepage acts as the main landing screen, featuring a **public calendar** where anyone can view upcoming events without needing to sign in. It also provides a secure login portal specifically for authorized users like **Club Coordinators**, the **General Secretary,** the **Students' Gymkhana President** and the **ADSA** to access their management dashboards.



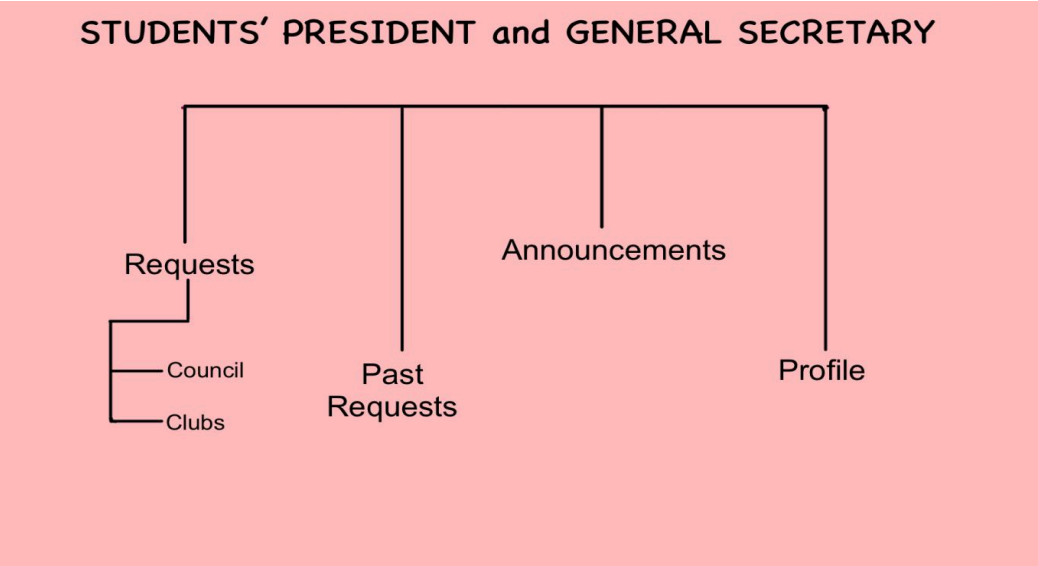
The web application supports five distinct login types, tailored to the specific roles involved in managing and approving campus events.



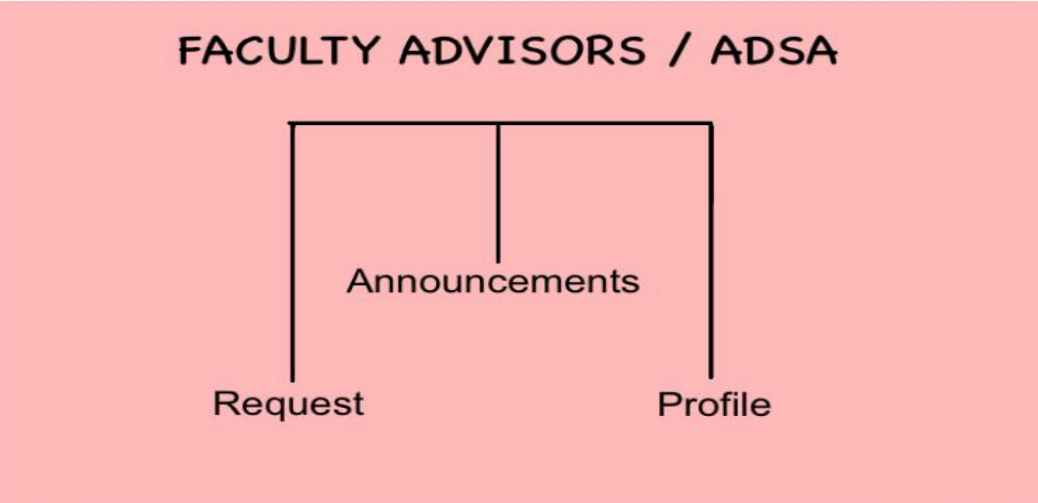
This will be the format of the **club coordinator's dashboard**. This will show the latest announcements and updates regarding budget reports. This will also allow the users to view their upcoming bookings updates as well as view their profile. There is an option for application submissions for upcoming campus events .



The below dashboard serves as the primary administrative interface for the **Students' President and General Secretary**. It features a centralized **Requests** module that allows leadership to manage incoming inquiries categorized by **Council** and **Clubs**. Additionally, the dashboard provides sections for tracking **Past Requests**, broadcasting campus-wide **Announcements**, and managing the executive's personal **Profile**.



The following dashboard serves as the primary administrative interface for **Faculty Advisors and the Associate Dean of Student Affairs (ADSA)**. It features a centralized **Request** module that allows faculty to manage and review incoming administrative inquiries. Additionally, the dashboard provides sections for broadcasting important **Announcements** and managing the user's personal **Profile**.



3.1.2. Hardware Interfaces

- A computing device for club coordinators and approving authorities to access the system via a web browser.
- Support for devices to receive One-Time Passwords (OTP) for secure approval of service requests
- Server-side hardware connected to the institutional intranet to host the web application and database.
- Network interface (Ethernet/Wi-Fi) to enable secure communication between clients and the server.

3.1.3. Software Interfaces

- The client-side components of the software must be able to operate seamlessly with various modern web browsers like Apple Safari 14+, Google Chrome 90+, Microsoft Edge 90+, Mozilla Firefox 90+.
- For ensuring authenticity of users the software shall verify credentials with the institute database.
- We will use multiple databases for storing the following: -
    a. User login credentials
    b. Upcoming Events
    c. Past service requests
    d. Past approved bills
    e. Current pending requests
    f. Popular Venues

3.2 Functional Requirements

3.2.1 Display of Public Calendar of All Events by Student Gymkhana

- The system shall display a public events calendar on the homepage, accessible to all users without login.
- The calendar shall display popular event venues such as LHC, OAT Stage, Auditorium, etc., along with the scheduled date and time of events.
- Users shall be able to view a brief summary of an event by hovering over the event entry on the calendar.
- Whenever a new event request is approved, the calendar shall automatically update by retrieving relevant details from the approved permission letter.

3.2.2 Seeking Permission for a New Event

On the Club Coordinator's homepage, the system shall provide the following service request options:

Permission Letter

- Selecting this option shall open a form that the user can fill and submit for approval.

Venue Booking

- The user shall be able to select the intended date and time for the event.
- The system shall display a list of available venues based on the selected slot.
- The user shall be able to fill out and submit the venue booking form.

MoUs (Memorandums of Understanding)

- The user shall be able to access and submit a request form for MoUs.

Bills and Receipts

- The user shall be able to upload approved and reimbursed bills and receipts.
- Uploaded expenses shall be deducted from the club's allocated budget.
- These transactions shall be stored in the past transactions archive.

Request Tracking

- Users shall be able to view the current status of submitted requests as **Pending**, **Approved**, or **Denied**.
- Users shall be able to view all past approvals along with their associated details.

3.2.3 Announcement Interface

Announcement Dashboard

- The system shall provide an Announcement tab within the profile dashboard of every Club Coordinator.
- This tab shall serve as a centralized hub for viewing all active and archived announcements.

Authority Access and Controls

- The system shall provide a "Send Announcement" action exclusively to authorized administrative roles:
    - Dean of Student Affairs (DOSA)
    - President Student Gymkhana (PSG)
    - General Secretary (GenSec)
    - Associate Dean of Student Affairs (ADSA)
    - Faculty Advisor
    - Financial Convenor
- The system shall log the sender's identity and timestamp for every announcement.

Announcement Composition and Distribution

- The system shall support:
    - Attachment of documents (PDFs or images).
    - Broadcasting announcements to all clubs simultaneously.
    - Filtering by specific club categories to send announcements.
- The system shall maintain an archive of past announcements for future reference.

3.2.4 Functionality for Easy Approval of Events

Event Approval Workflow

- The system shall support event approval by the following authorized roles:
    - Dean of Student Affairs (DOSA)
    - President Student Gymkhana
    - General Secretary of the concerned Club
    - Associate Dean of Student Affairs (ADSA)
    - Faculty Advisor
    - Financial Convenor
- Approvers shall receive an email notification whenever a new event request is created.
- Approvers shall be able to view a list of all pending requests, ordered by submission time.
- Approvers shall be able to select a request to:
    - View the event description
    - Review the complete submitted request form

Club Information Access

- Approvers shall be able to view the remaining budget of the respective club.
- Approvers shall be able to view the past bookings and event history of the club.

Digital Approval

- The system shall allow approvers to digitally sign and approve requests using OTP-based verification for security.

Request Rejection and Feedback

- Approvers shall have the option to reject an event request.
- When rejecting a request, the system shall require the approver to provide a reason for rejection.
- The rejection reason shall be stored and linked to the request for future reference.
- The concerned club/requestor shall be notified of the rejection along with the provided reason.
- Approvers shall be able to edit or update the rejection reason before final submission.

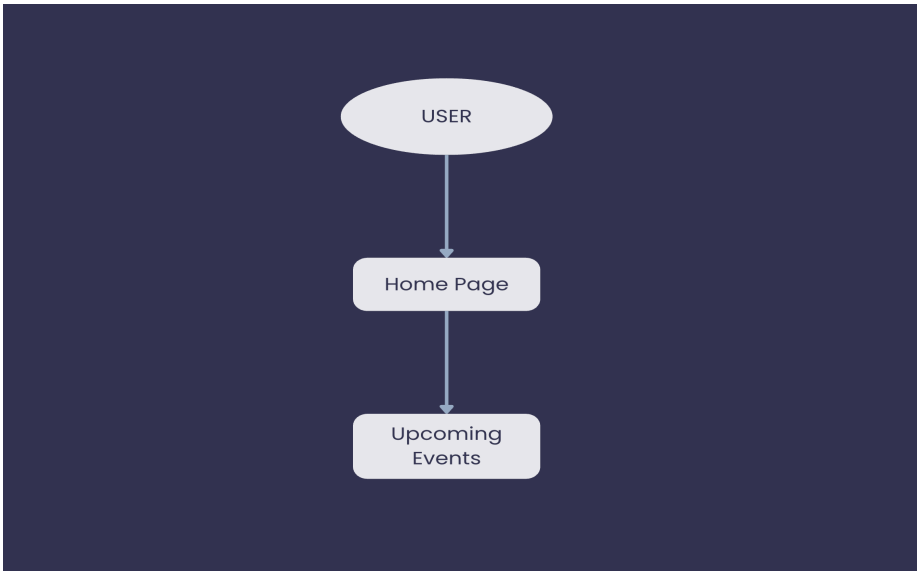
Audit Trail

- The system shall maintain an audit trail of all status changes, including the timestamp and approver identity.

3.2.5 Real-Time Budget Monitoring

- The system shall display the current club budget prominently on the Club Coordinator's homepage.
- The budget display shall include:
    - **Total Allocated Funds** – the initial amount granted for the fiscal period.
    - **Remaining Balance** – automatically updated based on approved expenditures.
    - **Expenditure Log** – a detailed record of all financial activity.
- The system shall allow coordinators to export financial summaries in **PDF** or **CSV** format for use in meetings or audits.

3.3. Use Case Model



3.3.1 Use Case #1 (Public Calendar)

Author – This use case was written by Harsh and Vanshika

Purpose - The system provides a centralized dashboard on the web app homepage that displays the public calendar, showing all upcoming events and activities organized by college clubs and student bodies.

Requirements Traceability – This feature fulfills the requirement for a unified campus activity feed and ensures that data from individual club schedules is aggregated and visible to the general student body in real-time.

Priority - Low

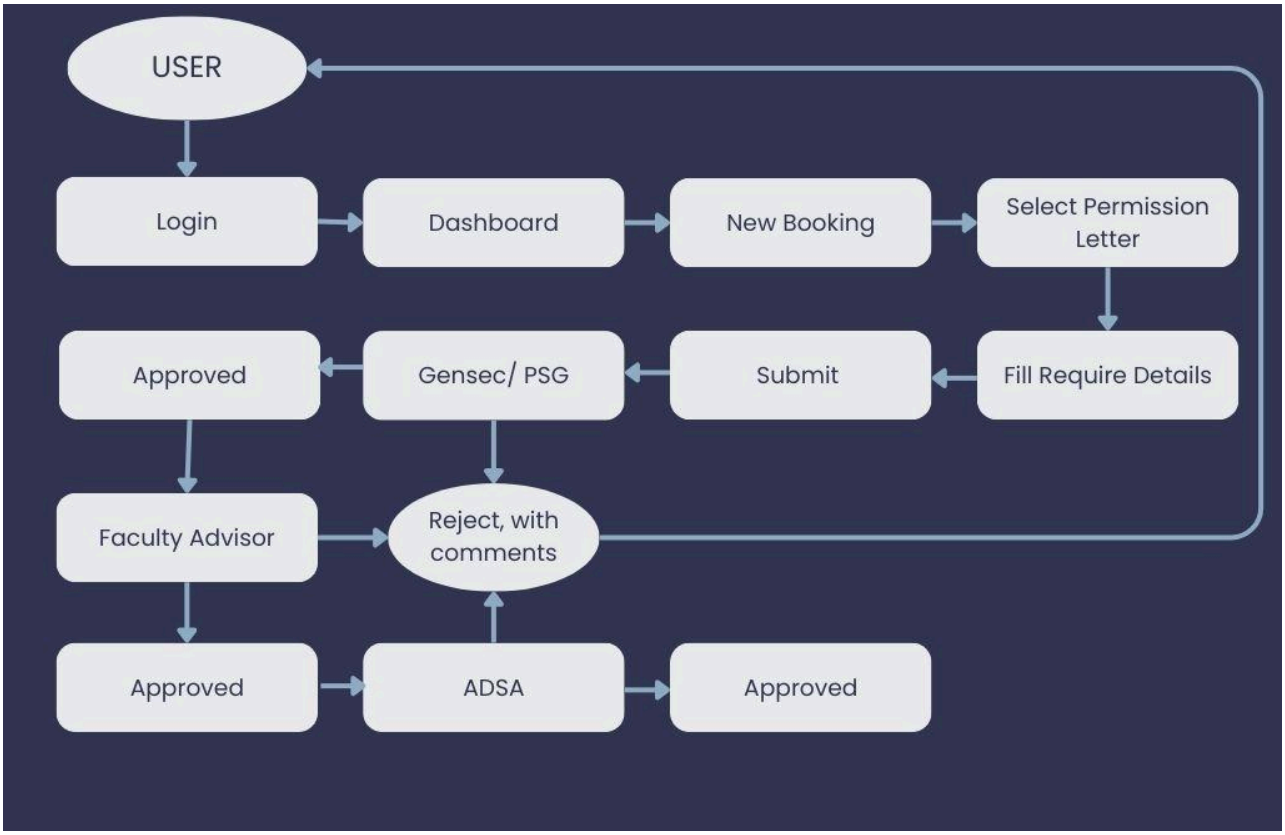
Preconditions - The database must be populated with events/activities of clubs, and the user must navigate to the web app homepage.

Post conditions - The user will be presented with a chronological view of upcoming events.

Actors – Actors in this use case are the students acting as the viewers and the Club Co-ordinators acting as the data source.

Exceptions -

- **Database Timeout:** If the system cannot retrieve event data, a "Service Temporarily Unavailable" message is displayed instead of the calendar.
- **Empty State:** If there are no upcoming events scheduled in the system, a placeholder graphic stating "No upcoming events" is shown to the user.



3.3.2 Use Case #2 (Permission Letter)

Author – This use case was written by Rishitesh and Sreeja.

Purpose -

To digitize the event permission process. Coordinators fill out a digital form and submit it for hierarchical approval. The system automates routing (GenSec → FacAd → ADSA) and allows coordinators to track real-time application status and history.

Requirements Traceability –

- The system shall provide a digital form for requesting permissions of events
- The system shall automatically route the application to the next authority in the fixed chain: General Secretary (GenSec) → Faculty Advisor (FacAd) → Associate Dean (ADSA).
- The system shall provide a dashboard for Coordinators to view real-time status (e.g., "Pending at FacAd") and historical records.

Priority - High

Preconditions -

- The user must be logged in as a registered Club Coordinator.
- The form is submitted in the proper given format.

Post conditions -

- A unique Application ID is generated for the permission request.
- The application status is updated to "Pending Approval" for the first authority (GenSec).
- The relevant authorities receive a notification to review the pending request.

Actors –

- Club Coordinator, General Secretary (GenSec), Faculty Advisor (FacAd), Associate Dean (ADSA), IITK Email (for OTP).

Exceptions -

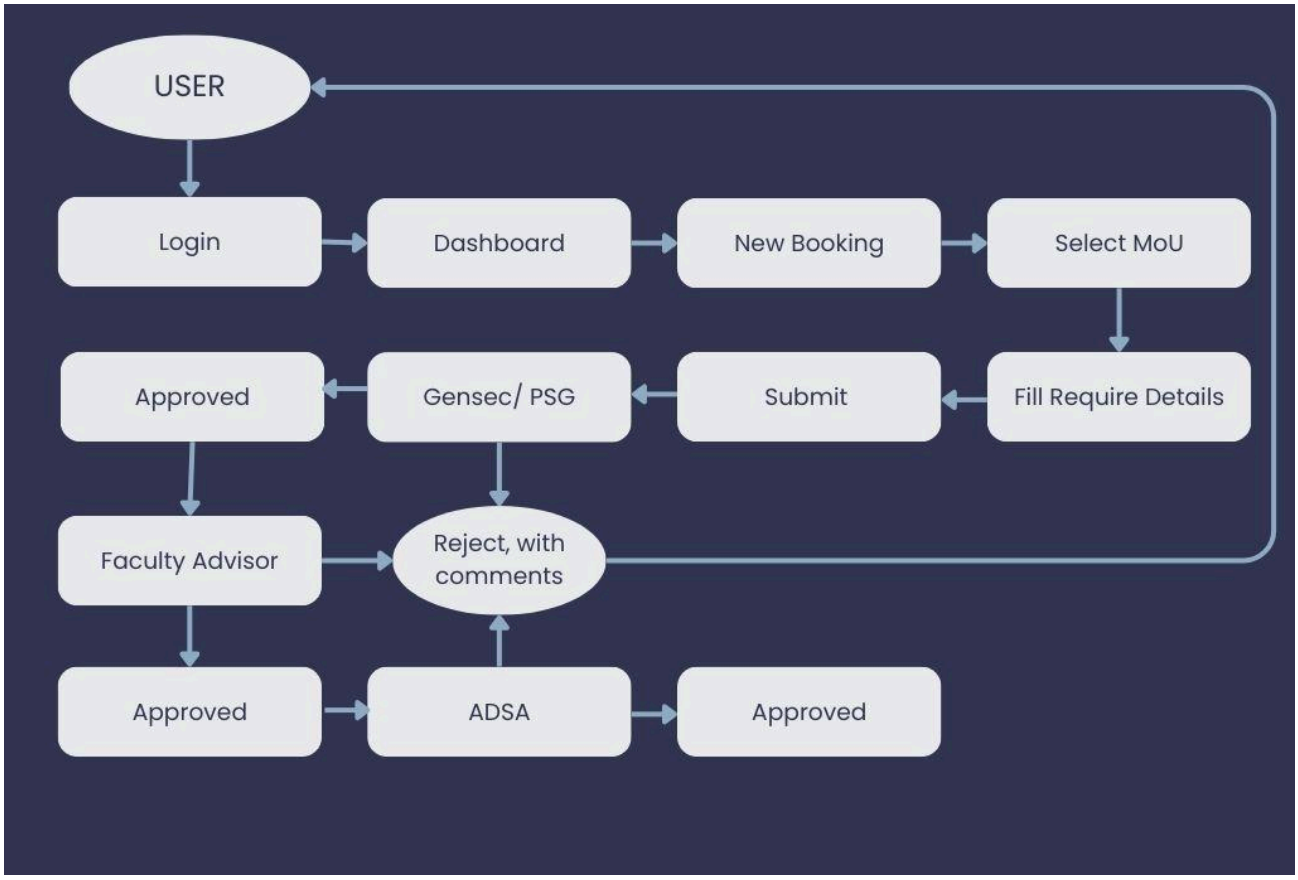
- If the entered OTP is incorrect, the digital signature fails, and the form submission is aborted.
- If any authority rejects the request, the workflow terminates immediately, the status is set to "Rejected," and the Coordinator is notified.
- If the OTP is not entered within the validity window, the session expires.

Includes

Public Calendar

Notes/Issues -

- The OTP verification serves as a binding digital signature for both the student representative and concerned authorities.



3.3.3 Use Case #3 (MoU)

Author – This use case was written by Vasu and Divya

Purpose -

To enable club coordinators to submit a Memorandum of Understanding (MoU) for inviting external individuals or organizations to campus, route it through the required approval authorities, and track its approval status.

Requirements Traceability –

- The system shall allow coordinators to submit an MoU form with required details and attachments.
- The system shall notify relevant authorities and support digital approval or rejection.
- The system shall forward the request through the defined approval workflow.
- The system shall allow coordinators to track approval status and pending signatories.

Priority - High

Preconditions -

- The user must be logged in and authenticated as a Club Coordinator.
- The club must be registered and active in the system.

Post conditions -

- An MoU request is created with a unique Request ID.
- Attached documents are stored and linked to the request.
- The MoU status is updated based on approval actions.
- Fully approved MoUs are marked as Approved and archived.

Actors –Club Coordinators, Approving Authorities (Council Head, PSG, Faculty Advisor, ADSA)

Exceptions -

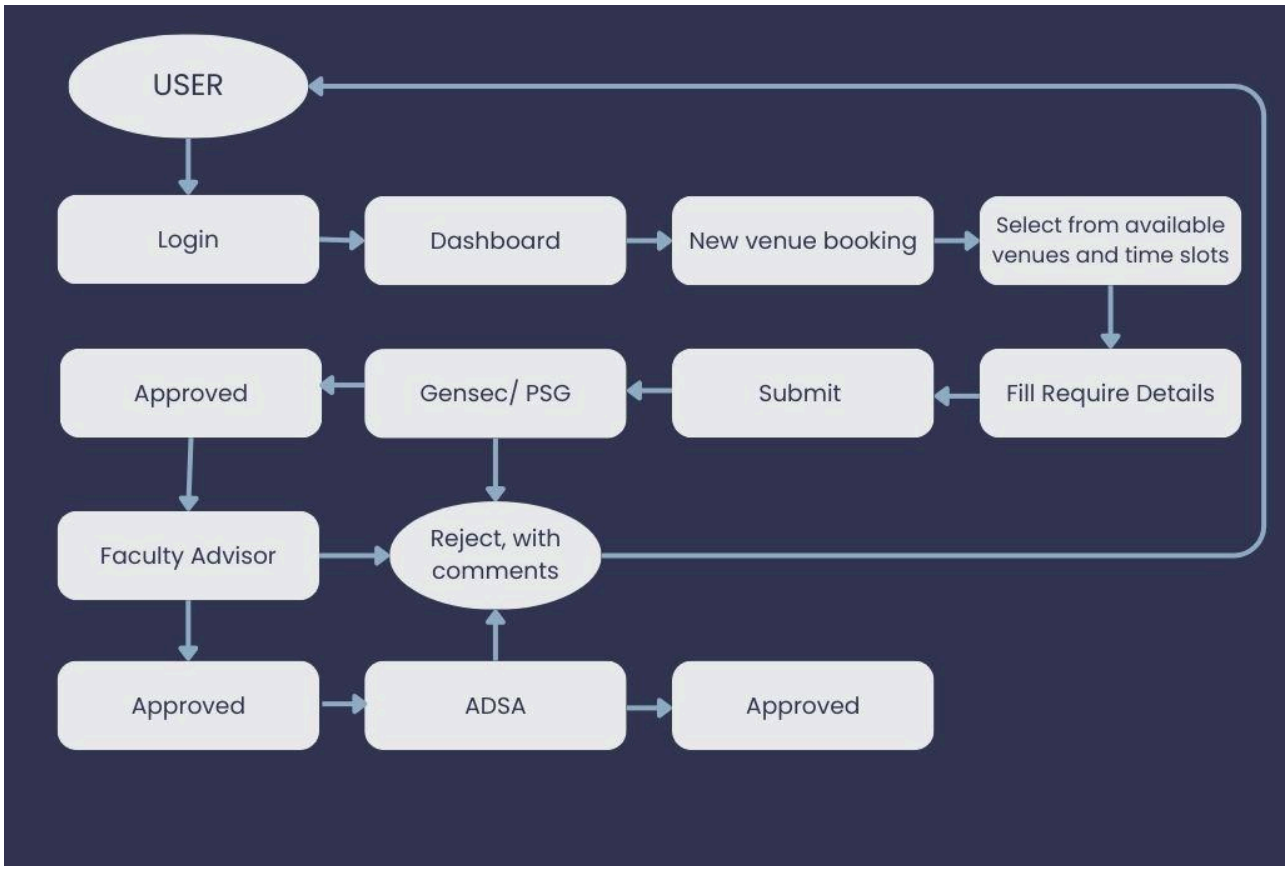
- Incomplete Submission: Submission is blocked if mandatory fields or attachments are missing.
- Rejection by Authority: The request is returned to the coordinator with reasons for modification and resubmission.

Includes -

Permission Letter

Notes/Issues-

Document formats should be standardized.



3.3.4 Use Case #4 (Venue Booking)

Author – This use case was written by Shruti and Saurya

Purpose –

- To enable club coordinators to request booking of institute venues for events, route their request through the required approval authorities, and track the approval status in real time.

Requirements Traceability –

- The system shall allow coordinators to initiate a venue booking request by selecting an intended date and day.
- The system shall display a list of available venues for the selected date and time.
- The system shall allow coordinators to submit a venue booking form with event and venue details.
- The system shall notify the relevant approving authorities upon form submission.
- The system shall allow authorities to approve or reject the request digitally.
- The system shall forward the request to the next required authority upon approval.

- The system shall allow coordinators to view the current approval status and pending signatories.

Priority – High

Preconditions –

- The user must be logged in and authenticated as a Club Coordinator.
- The club must be registered and active in the system.
- The venue availability data must exist in the system for the selected date.

Post conditions –

- A venue booking request is created and stored in the system with a unique Request ID.
- The approval status of the request is updated based on actions taken by authorities.
- Notifications are sent to the next required signatory after each approval.
- If fully approved, the venue is marked as booked for the specified date and time.

Actors – Club Coordinators, Approving Authorities (Council Head, PSG, Faculty Advisor, ADSA)

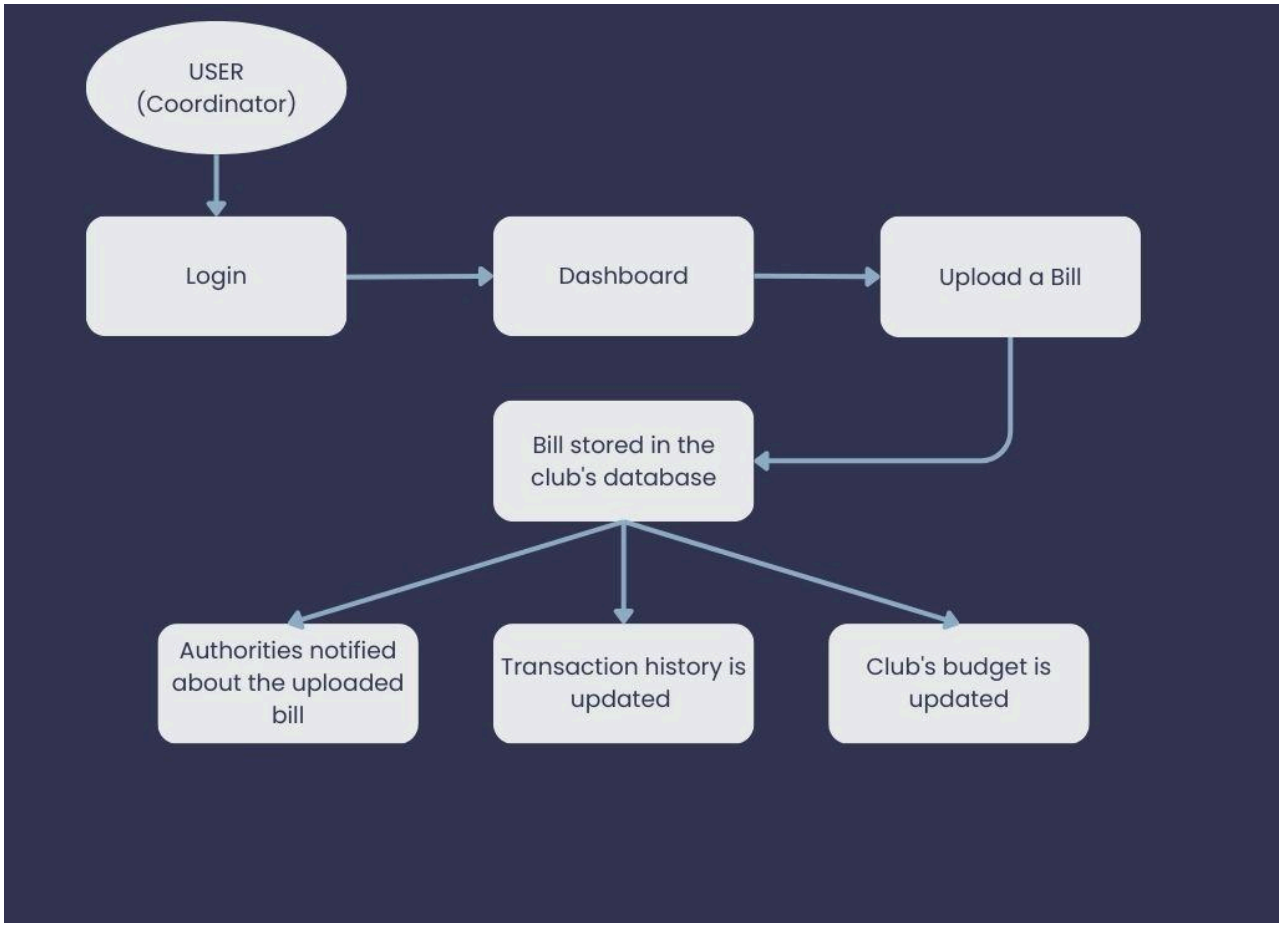
Exceptions –

- Venue Unavailable: The system prevents submission if the selected venue is already booked for the chosen date and time.
- Rejection by Authority: The request is sent back to the coordinator with the reason for rejection for modification and resubmission.

Includes – Permission Letter

Notes/Issues –

- Venue availability must be updated in real time to avoid double booking.
- The approval workflow may vary based on venue type and club category.



3.3.5 Use Case #5 (Bill Receipt)

Author – This use case was written by Diya and Pulkit

Purpose -

To allow club coordinators to submit their approved and reimbursed bill and receipts, so that an archive of their expenses is maintained and they can refer to their club's budget whenever required

Requirements Traceability –

- The system shall allow the users to submit their approved bills and receipts in pdf formats and also enter the amount of money that was reimbursed.
- The system will notify all the relevant authorities of the updation in the expenses of the concerned club
- The system will automatically deduct the amount from the club's current budget and display the updated budget in the coordinator's homepage
- The system will add this bill to the database of all the old transactions of the club

Priority - High

Preconditions -

- The user must be logged in and authenticated as a Club coordinator
- The status of the users account must be active
- The bill being submitted must have been pre-approved and reimbursed to the concerned student's account by the DOSA office

Post conditions -

- The club's current budget gets updated and the transaction history gets updated
- The relevant authorities are notified about the update in a club's transactions

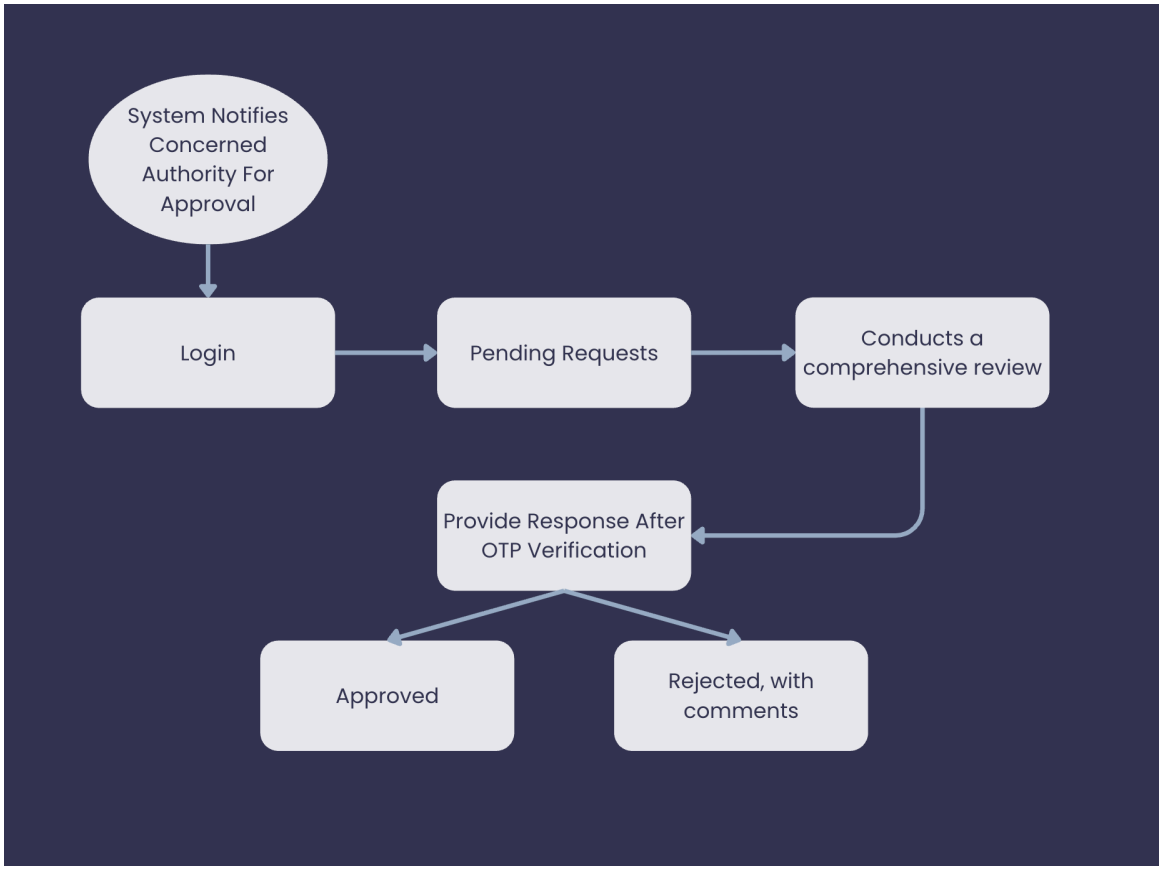
Actors – Club Coordinators, Approving Authorities (Council Head, PSG, Faculty Advisor, ADSA)

Exceptions -

- Non-approved bills are attached which will have to be rejected
- Bills and receipts do not match the amount entered in the system

Notes/Issues-

- Uploaded bills and receipts must be clearly legible and in supported formats
- Duplicate bill submissions should be detected and flagged by the system.



3.3.6 Use Case #6 (Approval of the Service Request)

Author – This use case was written by Saurya and Sreeja

Purpose -

Allow the authorities whose signatures are required to easily view and verify permission letters, venue booking requests and MoUs, digitally sign them and verify their signature using an OTP.

Requirements Traceability –

- The system shall send a submitted service request to the first concerned authority for approval
- The authority can digitally sign the form and verify using an OTP
- Once signed it goes to the next person in the approval pipeline
- If the form is rejected the user will be prompted to add a description of the reason for rejection
- The rejected form will be sent back to the club coordinator who can then modify the form and start with the approval process again

Priority - High

Preconditions -

- The approval authority must be logged in to their account and their account status must be active

Post conditions -

- Authorities will be able to verify the approval requests and accept or reject them
- The Club coordinators can change their form and resubmit in case rejected

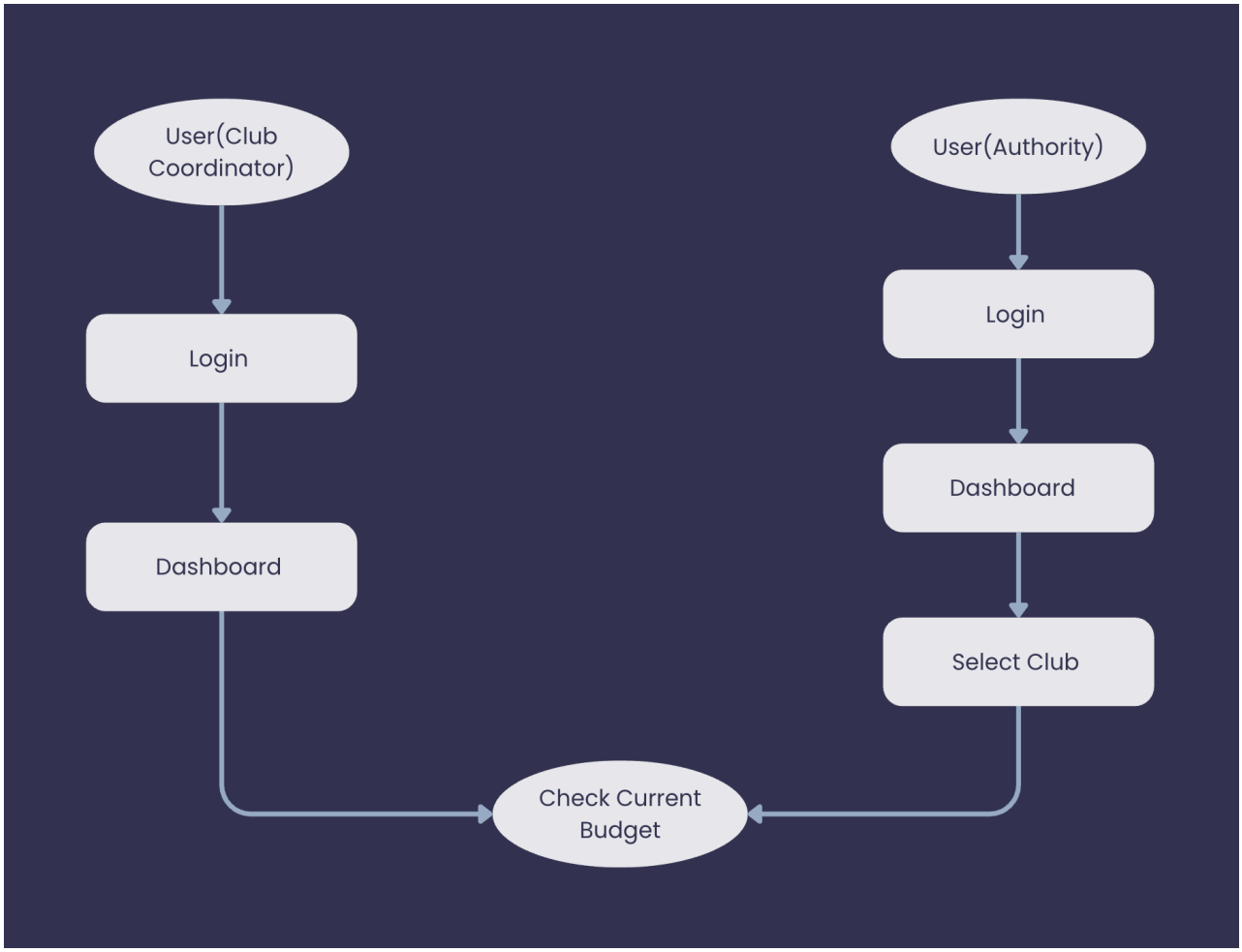
Actors – Club Coordinators, Approving Authorities (Council Head, PSG, Faculty Advisor, ADSA)

Exceptions -

- The approval action is rejected if the OTP is invalid or has expired.
- The system prevents approval actions if the user is not authorized for the current approval stage.

Notes/Issues -

- Approval actions must be final and non-editable once submitted.
- Rejection should require a mandatory reason to enable meaningful resubmission.
- The system should notify all relevant stakeholders on approval or rejection.



3.3.7 Use Case #7 (Look at Club Finances)

Author – This use case was written by Sreeja and Rishitesh.

Purpose -

To enable authorized stakeholders to view the club's budget balance, process reimbursement requests for expenses incurred by members, and maintain a detailed transaction log for auditing.

Requirements Traceability –

- The system shall show the coordinators the current budget left of the allocated budget
- The system shall provide a view for the GenSec and ADSA to monitor amount utilized vs. amount left.
- The system shall allow coordinators to input reimbursement requests based on submitted bills.
- The system shall automatically deduct approved reimbursements from the club balance.

- The system shall generate a timestamped transaction log for every financial activity (deduction, allocation, or reimbursement).

Priority - Medium

Preconditions -

- The user must be logged in and authenticated as a Coordinator, GenSec, ADSA or Faculty Advisor.
- The specific club must have an allocated budget for the current financial year existing in the database.

Post conditions -

- The club's balance is updated to reflect the deduction.
- The status of the specific bill/reimbursement is changed from Pending to Processed.
- A new entry is added to the Transaction Logs (containing Log ID, Amount, User ID, Timestamp, and Description).
- If a bill was processed, the balance is decreased by the exact bill amount, and a transaction log is created.

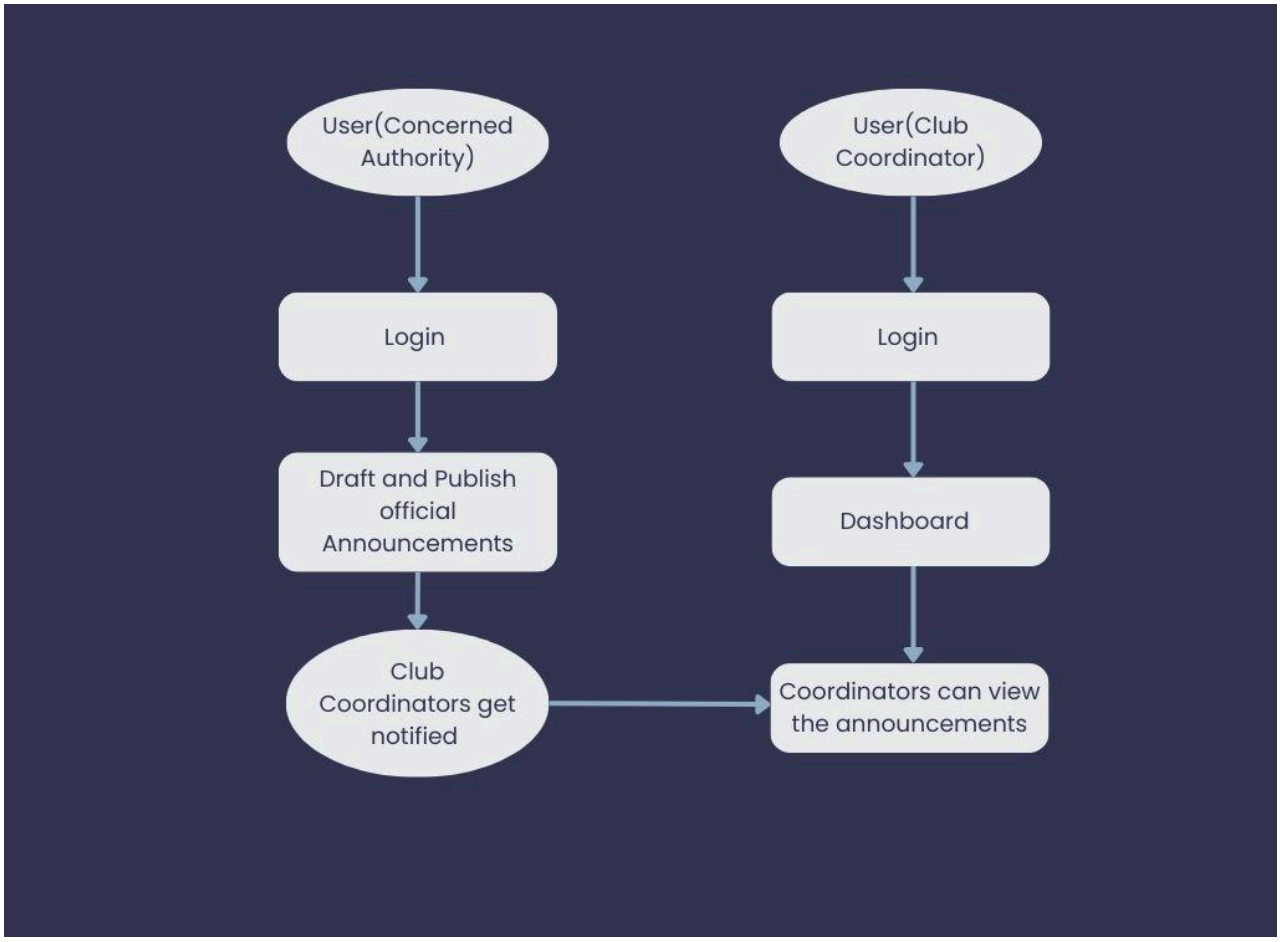
Actors – Club Coordinators, GenSec, ADSA

Includes -

- Bill Receipt

Notes/Issues -

- Transaction logs should be read-only for everyone to prevent tampering.



3.3.8 Use Case #8 (Announcement)

Author – This use case was written by Harsh

Purpose -

To empower authorities to communicate critical decisions and official updates to club coordinators via the dashboard's announcement tab.

Requirements Traceability –

- The system shall enable administrative authorities to draft and publish official announcements.
- Upon publication, the system shall automatically send an email notification to all relevant club coordinators.
- The system shall display the full announcement within a dedicated section of the Club Coordinator's dashboard.

Priority - Medium

Preconditions -

- **To Publish:** The user must be logged in and authenticated with **GenSec**, **ADSA**, or **Faculty Advisor** privileges.
- **To View:** The **Club Coordinator** must be logged in to the system to access the dashboard notifications.

Post conditions -

- The announcement data is stored in the database, and the system updates the dashboard interface for all Club Coordinators to display the new message.

Actors – Club Coordinators, GenSec, ADSA, Faculty Advisor

Notes/Issues -

- To ensure official authority, only designated administrators can publish announcements; Club Coordinators typically have read-only access to this feature.

4. Other Non-functional Requirements

4.1. Performance Requirements

- **Concurrency** – The platform must support 150 concurrent users without performance degradation to ensure uninterrupted workflow during peak submission periods.
- **Responsiveness** – Dashboard loading times shall remain below 2 seconds, while queries for the "Interactive Venue Scheduler" must return results in under 1 second to prevent booking conflicts.
- **Real-time Synchronization** – Status updates (e.g., from "Pending" to "Approved") must be reflected on the Coordinator's dashboard within 5 seconds of the Authority's action.
- **Resource Efficiency** – The system shall utilize optimized database indexing to ensure that retrieving archival data does not slow down active daily operations.

4.2. Safety and Security Requirements

Identity Authentication

- The system must use the Institute's existing LDAP/Webmail for login. We will not store any passwords on our own servers. This is critical to prevent credential theft; if our database is compromised, student email accounts remain safe.
- **Session Timeouts:** User sessions must expire after 15 minutes of inactivity. This is a necessary safeguard against "walk-away" attacks where a student might leave their account open on a public library computer, allowing others to access it.

Prevention of Financial Fraud (Fake Bills)

- **Immutable Transaction Logs:** To prevent the submission of fake bills or the alteration of reimbursement amounts after approval, the system must use an "append-only" database strategy for financial records. Once a bill amount is entered and approved, it cannot be edited—only a counter-transaction can correct it.
- **Duplicate Detection:** The system must implement a check to flag if a bill with the exact same invoice number and amount has been uploaded previously, preventing double-counting or re-submission of old receipts.

Access Control & Anti-Impersonation

- **Strict Role Separation:** We must ensure that a user logged in as a "Student" can never access the "Approver" API endpoints. This prevents a scenario where a tech-savvy student might try to force-approve their own event by manipulating the browser code.
- **OTP Verification for Critical Actions:** A simple click is not enough for sensitive actions. Approving a budget release or signing a permission letter must require a One-Time Password (OTP) sent to the Faculty or ADSA's official email. This ensures that even if a device is left unlocked, an unauthorized person cannot digitally sign a legal document.

Data Availability & Conflict Prevention

- **Automated Backups:** Since the DOSA office handles critical semester-long data, the system must perform encrypted backups every 24 hours. This safeguards against data loss due to server crashes or accidental deletion by admins.
- **Prevention of Simultaneous Bookings (Race Conditions):** The system must handle "race conditions" where two different coordinators try to book the same Lecture Hall for the same time slot at the exact same second. The database must lock the slot for the first request that hits the server, ensuring that double bookings are physically impossible.

4.3. Software Quality Attributes

4.3.1 Usability

- **Workflow Efficiency:** To solve the bottleneck of "unavailable signatories," the approval interface must allow authorities to review and digitally sign a proposal in fewer than **3 clicks**.
- **Visual Clarity:** The Interactive Venue Scheduler shall use distinct color coding (Red/Green) to ensure coordinators can identify and select conflict-free slots in under **5 seconds**.
- **Input Validation:** The system must validate forms in real-time (e.g., blocking negative budget values) to prevent the submission of malformed proposals.

4.3.2 Reliability

- **Conflict Prevention:** The system must enforce strict database transactions to ensure that if two users attempt to book the same venue simultaneously, only **one** booking is confirmed.
- **Data Consistency:** To ensure transparency, the Financial Dashboard must reflect approved reimbursements immediately, updating the "Remaining Budget" value within **2 seconds** of approval.

4.3.3 Maintainability

- **Modularity:** The codebase will utilize a strict separation between Frontend (React) and Backend (API) logic, ensuring that UI updates do not break core approval workflows.
- **Configurability:** Variable administrative rules (e.g., Budget Caps, Current Semester Dates) must be stored in a central database table, allowing updates without modifying source code.

4.3.4 Portability

- **Platform Independence:** The system is a **Desktop Web Application** designed to function identically on Chrome, Edge, and Firefox across Windows, macOS, and Linux without requiring plugins.

Appendix A – Data Dictionary

1. User Class

Description	Base class for all users. Stores login data for both Club Coordinators and Authorities.
Class Variables	name: string, iitk_email: string, user_id: string, role: enum {Coordinator, GenSec, PSG, FacAd, ADSA}, Password_hash: string
Methods	login(), logout(), view_public_calendar()
Relationships	Parent class for **Club Coordinator** and **Authority**.

2. Club Coordinator Class

Description	Represents student executives responsible for raising requests.
Class Variables	club_name: string, budget_allocated: float, budget_remaining: float, pending_bookings: int
Methods	draft_proposal, submit_bill, check_budget
Relationships	Subclass of **User**. Has one-to-many relationship with **Service Request**.

3. Authority Class

Description	Roles with approval powers (Faculty, DOSA, Gymkhana President).
Class variables	designation: string, digital_signature_token: string, otp_secret: string
Methods	verify_otp, approve_request, reject_request
Relationships	Subclass of **User**. Approves **Service Requests**.

4. Service Request Class

Description	Represents any formal request (Venue Booking, Permission, or MoU).
Class variables	request_id: string, type: enum, status: enum, event_date: datetime, is_public: bool
Methods	submit, update_status, get_public_details
Relationships	Linked to **Venue** and created by **Club Coordinator**.

5. Venue Class

Description	Stores data about locations like Lecture Halls or OAT.
Class variables	venue_id: string, name: string, booked_slots: list
Methods	check_availability, lock_slot
Relationships	Used by **Service Request** to prevent double-booking.

6. Financial Transaction Class

Description	Represents budget changes and bill reimbursements.
Class variables	transaction_id: string, club_id: string, amount: float, invoice_number: string
Methods	record_transaction, check_duplicate_invoice
Relationships	Updates **Club Coordinator** budget. Stores uploaded bills.

7. Announcement Class

Description	Official updates sent by authorities to coordinators.
Class variables	announcement_id: string, sender_id: string, message: text, target_clubs: list
Methods	publish
Relationships	Created by **Authority**. Viewed by **Club Coordinator**.

Appendix B - Group Log

Date	Timings	Durations	Minutes
07 Jan	11:00 – 12:30	1.5 hrs	Idea Suggestions by everyone. Main Ideas proposed and presented were: • Second Hand Goods and Rental Service • Carpooling System • Gymkhana Automation System
08 Jan	11:00 – 13:00	2 hrs	• Preliminary discussion on feasibility and usefulness of ideas. • The administrative framework of student bodies currently suffers from inefficiencies caused by a heavy reliance on manual, paper-based workflows. of a unified system to manage the same. • Decided to implement a software solution for the same. • Finalized Gymkhana Automation System.
09 Jan	18:00 – 20:00	2 hrs	• Finalized the description of the project and named the project as “Swifty”. • Consulted with faculty to validate a startup concept, focusing on technical feasibility and market scalability.
10 Jan	18:00 – 20:00	3 hrs	• Started work on the SRS document. • Discussed UI and functional requirements.
14 Jan	16:00 – 17:00	1 hrs	• We discussed the various frameworks to be used in the frontend and backend of this software.
16 Jan	18:30 – 21:00	2.5 hrs	• Formalized the features of the software. • Exploration of possible implementation, challenges. • Divided the various sections of the SRS document amongst ourselves.
17 Jan	18:00 – 21:00	3 hrs	• Raised and settled contentions regarding various use cases, features and the algorithm to be used in the software. • Finalised the main features of pair-matching and slot-booking systems and documented it in the SRS.
19 Jan	18:00 – 21:00	3 hrs	• Created the first draft of the SRS document.

21 Jan	21:00 – 23:00	2 hrs	• Proofread the whole document.